

Student Mental Health Monitoring System

SE LAB Experiment-05 BATCH-D

Estimation of Software Development Effort and Cost
using Use Case Point (UCP) Method

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1 MODIFIED USE CASE DIAGRAM

The diagram below annotates every actor and use case with its UCP complexity classification: S = Simple, A = Average, C = Complex.

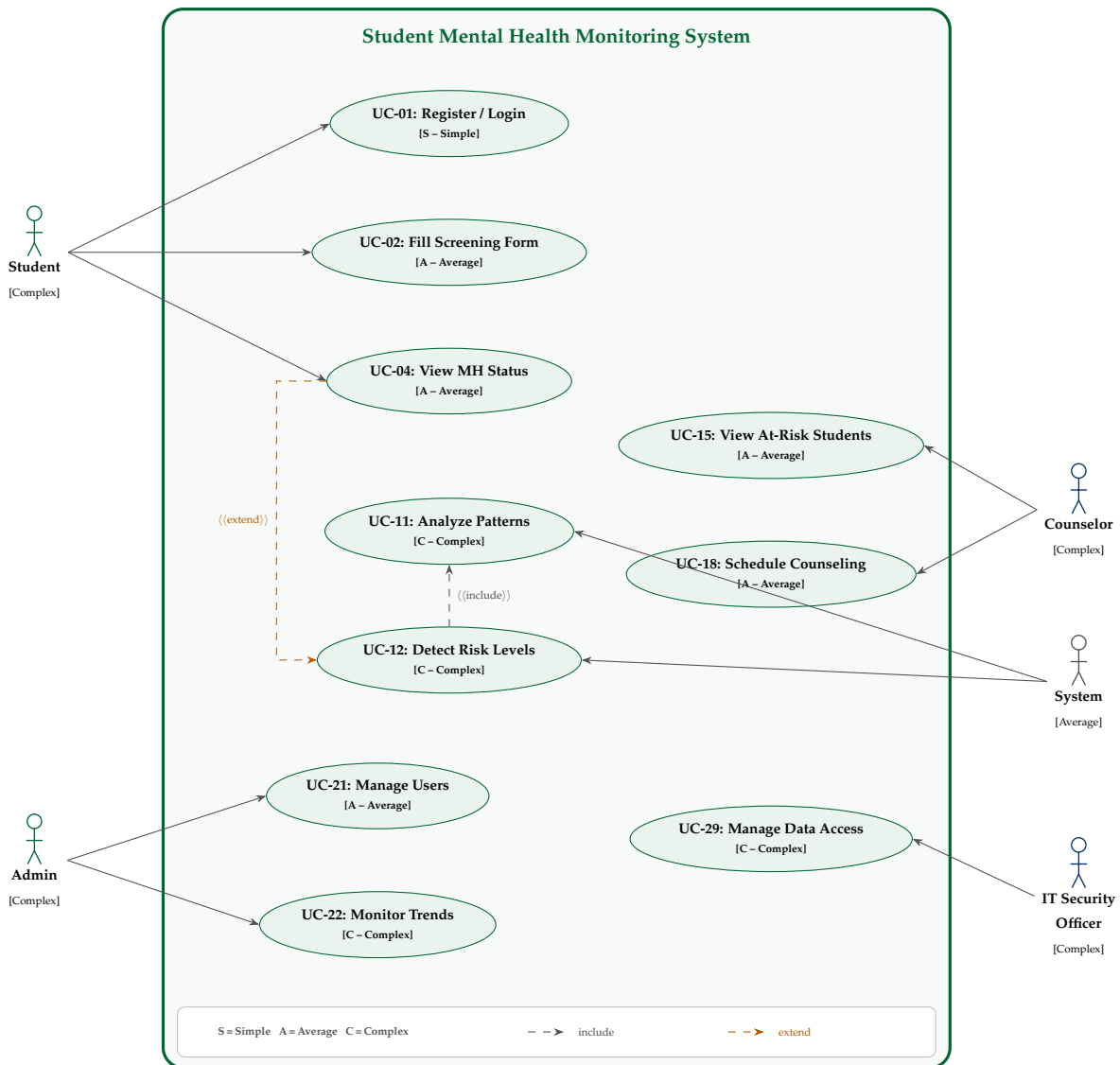


Figure 1: Modified Use Case Diagram with UCP Complexity Annotations

2 ACTOR CLASSIFICATION TABLE

Actor Complexity Rules: Simple (1) = API system. Average (2) = Protocol/interface. Complex (3) = Human with full GUI.

#	Actor	Complexity	Weight	Justification
1	Student	Complex	3	Human user interacting via full web/mobile GUI (login, screening forms, dashboard)
2	Counselor	Complex	3	Human user with rich GUI: views at-risk lists, schedules sessions, reads reports
3	Admin	Complex	3	Human user managing users, roles and institutional trend reports via GUI
4	IT Security Officer	Complex	3	Human user configuring policies and reviewing audit logs through dedicated UI
5	System (Scheduler / Analytics)	Average	2	Internal automated component invoked by scheduler; uses internal API, no GUI
Unadjusted Actor Weight (UAW)				$(4 \times 3) + (1 \times 2) = 14$

Table 1: Actor Classification and Unadjusted Actor Weight (UAW)

3 USE CASE CLASSIFICATION TABLE

Use Case Complexity Rules: Simple (5) = ≤ 3 transactions. Average (10) = 4–7 transactions. Complex (15) = > 7 transactions.

UC-ID	Use Case	Complexity	Wt	Main Trans.	Alt Flows	Justification
UC-01	Register / Login	Simple	5	3	1	Enter credentials, validate, redirect 1 alt for failed login
UC-02	Fill Screening Form	Average	10	5	2	Load, fill, validate, save, score; alts for incomplete & timeout
UC-04	View MH Status	Average	10	4	1	Fetch record, compute risk, render; alt for no-data state
UC-11	Analyze Patterns	Complex	15	8	2	Trigger, load, normalise, detect, store, display; alts for insufficient data
UC-12	Detect Risk Levels	Complex	15	9	2	Receive, threshold, classify (3 tiers) save, flag, notify, display
UC-15	View At-Risk Students	Average	10	5	1	Login, open panel, fetch, render list optionally schedule
UC-18	Schedule Counseling	Average	10	7	2	Select slot, check, create session, notify, confirm; alts for unavail & cancel
UC-21	Manage Users	Average	10	6	1	CRUD on profiles, role assignment alt for invalid input
UC-22	Monitor Trends	Complex	15	8	1	Aggregate, generate report, render charts, export; alt for no data
UC-29	Manage Data Access	Complex	15	9	2	Configure policy, apply rules, validate, log, audit; alts for non-compliance
Totals: S=1, A=5, C=4			S:1 × 5 = 5 A:5 × 10 = 50 C:4 × 15 = 60 UUCW = 115			

Table 2: Use Case Classification and Unadjusted Use Case Weight (UUCW)

4 UNADJUSTED USE CASE POINTS (UUCP)

$$\text{UUCP} = \text{UAW} + \text{UUCW} = 14 + 115 = 129$$

Component	Calculation	Sub-total
Unadjusted Actor Weight (UAW)		14
Complex Actors (4×3)	4×3	12
Average Actors (1×2)	1×2	2
Unadjusted Use Case Weight (UUCW)		115
Simple UCs (1×5)	1×5	5
Average UCs (5×10)	5×10	50
Complex UCs (4×15)	4×15	60
UUCP = UAW + UUCW	14+115	129

Table 3: UUCP Computation Summary

5 TECHNICAL COMPLEXITY FACTOR (TCF)

$$\text{TCF} = 0.6 + (0.01 \times \sum_{i=1}^{13} W_i \times V_i) \quad W_i = \text{standard weight}, \quad V_i \in [0,5]$$

T#	Technical Factor	W_i	V_i	$W_i \times V_i$	Justification
T1	Distributed system	2	2	4	Client-server; moderate distribution
T2	Response time / performance	1	4	4	Real-time risk detection needs fast response
T3	End-user efficiency	1	4	4	UI must be intuitive for non-technical students
T4	Complex internal processing	1	4	4	ML-based pattern analysis and risk classification
T5	Reusability of code	1	3	3	Moderate reuse: shared auth, dashboard widgets
T6	Installation ease	0.5	2	1	Institutional deployment; some setup required
T7	Operational ease	0.5	4	2	Must be accessible with no IT background
T8	Portability	2	3	6	Web + mobile access required
T9	Changeability / maintainability	1	3	3	Moderate; modular design planned
T10	Concurrency / multiple users	1	4	4	Hundreds of students may access simultaneously
T11	Special security features	1	5	5	Sensitive health data; GDPR-like compliance
T12	Direct third-party access	1	1	1	No significant third-party integration planned
T13	Special user training	1	2	2	Minimal training; counselors need brief orientation
Total $\sum W_i \times V_i$				43	

Table 4: Technical Complexity Factor – Calculation Sheet

$$\text{TCF} = 0.6 + (0.01 \times 43) = 0.6 + 0.43 = \mathbf{1.03}$$

Note: TCF = 1.03 (slightly above average). Key drivers: T11 (security), T10 (concurrency), T8 (portability), T4 (ML processing). Valid range: 0.6–1.3.

6 ENVIRONMENTAL FACTOR (EF)

$$EF = 1.4 + (-0.03 \times \sum_{i=1}^8 W_i \times V_i)$$

E1–E6: higher V_i reduces effort. E7–E8: weights set to 0 (non-negative constraint applied).

E#	Environmental Factor	W_i	V_i	$W_i \times V_i$	Justification
E1	Familiarity with UML / dev process	1.5	3	4.5	Moderate; studied SE methods in coursework
E2	Application experience	0.5	2	1.0	First time building a full healthcare system
E3	Object-oriented experience	1.0	3	3.0	Good OOP background from prior courses
E4	Lead analyst capability	0.5	3	1.5	Academically competent; limited industry exposure
E5	Motivation of team	1.0	4	4.0	High motivation; personal interest in project
E6	Stable requirements	2.0	3	6.0	Requirements moderately stable; scope defined
E7	Part-time staff	0	3	0	Weight set to zero (non-negative constraint)
E8	Difficult programming language	0	2	0	Weight set to zero (non-negative constraint)
Total $\sum W_i \times V_i$				20.0	

Table 5: Environmental Factor – Calculation Sheet

$$EF = 1.4 + (-0.03 \times 20.0) = 1.4 - 0.60 = \mathbf{0.80}$$

Note: EF = 0.80 (below 1.0) indicates a favourable team environment that reduces effort. Driven by high motivation (E5), stable requirements (E6), and OOP experience (E3). Negative weights for E7 and E8 are set to zero per non-negative constraint.

7 USE CASE POINTS (UCP) CALCULATION

$$\text{UCP} = \text{UUCP} \times \text{TCF} \times \text{EF}$$

Parameter	Formula	Value
Unadjusted Use Case Points (UUCP)	$14 + 115$	129
Technical Complexity Factor (TCF)	$0.6 + 0.01 \times 43$	1.03
Environmental Factor (EF)	$1.4 - 0.03 \times 20.0$	0.80
Use Case Points (UCP)	$129 \times 1.03 \times 0.80$	106.30

Table 6: Use Case Points (UCP) Computation

$$\text{UCP} = 129 \times 1.03 \times 0.80 = 106.30$$

8 EFFORT ESTIMATION

$$\text{Effort} = \text{UCP} \times \text{PF}$$

Productivity Factor: **28 person-hours / UCP** (domain-adjusted for healthcare system complexity and real-world academic project environment).

Component	Calculation	Result
Use Case Points (UCP)	–	106.30
Productivity Factor (PF)	Domain-adjusted	28 hrs / UCP
Total Effort	106.30×28	2,976.4 person-hours
Team size	2 developers	2 persons
Working hours / day	–	6 hours
Working days / week	–	5 days
Effort per person	$2976.4 \div 2$	1,488.2 hrs / person
Duration (weeks)	$1488.2 \div (6 \times 5)$	≈ 50 weeks
Duration (months)	$50 \div 4.33$	≈ 11.5 months

Table 7: Effort Estimation Summary

Total Effort = $106.30 \times 28 = 2,976.4$ person-hours (≈ 50 weeks for a 2-person team)

9 FINAL COST ESTIMATION

$$\text{Cost} = \text{Effort} \times \text{Rate}$$

Assumed rate: **Rs. 900 / person-hour** (mid-level developer rate, India).

Cost Component	Calculation	Amount
Development Effort Cost	$2976.4 \times \text{Rs. } 900$	Rs. 26,78,760
Contingency (15%)	$26,78,760 \times 0.15$	Rs. 4,01,814
Infrastructure / Tools (5%)	$26,78,760 \times 0.05$	Rs. 1,33,938
Testing & QA (10%)	$26,78,760 \times 0.10$	Rs. 2,67,876
Total Project Cost	Sum of all above	Rs. 34,82,388

Table 8: Final Project Cost Estimation

Total Estimated Project Cost = Rs. 34,82,388 ($\approx$ Rs. 34.8 Lakhs)

10 ANALYSIS OF ESTIMATION SENSITIVITY

10.1 Impact of Increasing Technical Complexity

Scenario: All T-factors raised to maximum (5), increasing $\sum W_i \times V_i$ from 43 to 65.

Parameter	Baseline	High Complexity	Change
$\sum W_i \times V_i$	43	65	+22
TCF	1.03	$0.6 + 0.01 \times 65 = 1.25$	+0.22
UCP	106.30	$129 \times 1.25 \times 0.80 = 129.00$	+22.70
Effort (hrs)	2,976.4	$129.00 \times 28 = 3,612.0$	+635.6
Cost (Rs.)	26,78,760	32,50,800	+5,72,040
Cost increase	–	–	+21.4%

Table 9: Sensitivity: Maximum Technical Complexity

Finding: Max technical complexity raises cost by **+21.4%** ($\approx$ Rs. 5.7L extra). Key drivers: T11 (security) and T4 (ML processing). **Mitigation:** Use proven frameworks (scikit-learn, TensorFlow).

10.2 Impact of Low Team Experience

Scenario: E1–E6 drop by 1 point each, reducing $\sum W_i \times V_i$ from 20.0 to 13.5.

Parameter	Baseline	Low Experience	Change
$\sum W_i \times V_i$ (EF)	20.0	13.5	-6.5
EF	0.80	$1.4 - 0.03 \times 13.5 = 0.995$	+0.195
UCP	106.30	$129 \times 1.03 \times 0.995 = 132.27$	+25.97
Effort (hrs)	2,976.4	$132.27 \times 28 = 3,703.6$	+727.2
Cost (Rs.)	26,78,760	33,33,240	+6,54,480
Cost increase	–	–	+24.4%

Table 10: Sensitivity: Low Team Experience

Finding: Low experience inflates cost by **+24.4%** ($\approx$ Rs. 6.5L). Critical gaps: ML and security. **Mitigation:** 2-week learning sprint before development; use high-level libraries.

10.3 Sensitivity if Productivity Factor Changes (PF: 28 $\rightarrow$ 35 hrs/UCP)

Scenario: PF increases from 28 to 35 hrs/UCP (higher-than-expected complexity). All other values at baseline.

Parameter	PF = 28	PF = 35	Change
UCP	106.30	106.30	–
Productivity Factor (hrs/UCP)	28	35	+7
Effort (person-hours)	2,976.4	$106.30 \times 35 = 3,720.5$	+744.1
Duration (weeks, 2 persons)	49.6	$3720.5 \div 60 = 62.0$	+12.4
Development Cost (Rs.)	26,78,760	33,48,450	+6,69,690
Cost increase	–	–	+25.0%

Table 11: Sensitivity: Productivity Factor Change (PF 28 $\rightarrow$ 35)

Finding: A 25% rise in PF causes a **linear +25% cost increase**. Most likely cause: under-estimating analytics engine and security subsystem. **Mitigation:** Budget at PF = 30; track Agile sprint velocity.

10.4 Combined Sensitivity Summary

Scenario	Effort (hrs)	Cost (Rs.)	% Change
Baseline (TCF=1.03, EF=0.80, PF=28)	2,976.4	26,78,760	–
High Technical Complexity (TCF=1.25)	3,612.0	32,50,800	+21.4%
Low Team Experience (EF=0.995)	3,703.6	33,33,240	+24.4%
Higher PF (PF=35)	3,720.5	33,48,450	+25.0%
Worst Case (all three)	5,614	50,52,600	+88%

Table 12: Combined Sensitivity Analysis Summary

Worst-Case Calculation:

$$UCP_{\text{worst}} = 129 \times 1.25 \times 0.995 \approx 160.4$$

$$\text{Effort}_{\text{worst}} = 160.4 \times 35 \approx 5,614 \text{ hrs}$$

$$\text{Cost}_{\text{worst}} = 5,614 \times 900 \approx \text{Rs. } 50,52,600$$

Key Insights:

1. Baseline Rs. 34.8 Lakhs can rise to Rs. 50+ Lakhs under worst-case conditions.
2. Productivity factor is the most sensitive lever (linear cost impact).
3. Use Agile velocity tracking to detect deviations early.
4. Set **recommended budget = Rs. 42 Lakhs** (20% buffer over baseline).

11 ANALYSIS OF ESTIMATION ACCURACY

11.1 Strengths of the UCP Estimate

#	Strength	Evidence in SMHMS
1	Scope-driven (use-case based)	All 10 UCs are well-defined with clear transaction counts
2	Technical environment captured	TCF captures SMHMS concerns: security, concurrency, portability
3	Team environment captured	EF reflects 2-person academic team with experience gaps
4	Traceable & auditable	Every parameter is documented, reviewable and reproducible
5	Sensitivity quantified	Three scenarios bound cost range: Rs. 27L – Rs. 50L

Table 13: Strengths of the UCP Estimation

11.2 Limitations and Sources of Inaccuracy

#	Limitation	Impact on SMHMS	Mitigation
1	Subjectivity in V_i ratings	TCF/EF depend on judgment; raters may differ	Peer-review; average across team
2	Domain-adjusted PF	PF=28 is estimated; healthcare ML may need more	Use PF=30 for budget buffer
3	No non-functional effort	UCP ignores docs, testing overhead	Add 25% overhead
4	Transaction count variability	UC-11/UC-12 counts debatable	Cross-validate with Lab-04 sequence diagrams
5	No historical calibration	First project of this type	Track actual vs estimated; recalibrate PF

Table 14: Limitations and Mitigation Strategies

11.3 Final Estimation Summary

Metric	Value
UUCP	129
TCF	1.03
EF	0.80
UCP	106.30
Productivity Factor	28 hrs / UCP
Total Effort	2,976.4 person-hours
Duration (2-person team)	≈50 weeks (≈11.5 months)
Hourly Rate	Rs. 900 / person-hour
Development Cost	Rs. 26,78,760
Total Cost (with overheads)	Rs. 34,82,388 (≈Rs. 34.8 Lakhs)
Recommended Budget	Rs. 42,00,000 (20% buffer)
Worst-Case Cost Cap	Rs. ≈ 50,00,000

Table 15: Final UCP Estimation Summary for SMHMS

Baseline: Rs. 34,82,388 | Recommended Budget: Rs. 42,00,000 | Worst-Case Cap: Rs. 50,00,000